

Design Project Proposal Form

WINDOW COMPARATOR USING IC741 AND TWO RED LEDs

Submitted by

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In partial fulfilment of the course

21ECC222L

ANALOG AND LINEAR ELECTRONIC CIRCUITS LABORATORY



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

SRM Institute of Science and Technology (Deemed University)

(Under Section 3 of the UGC Act, 1956)

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Title

Window Comparator Circuit Using Two IC741 Op-Amps and Two Red LEDs for Input Voltage Window Detection

Abstract / Problem Statement

This project presents the design and implementation of a Window Comparator circuit using two IC741 operational amplifiers configured as open-loop voltage comparators, with two red LEDs as output status indicators. A window comparator detects whether an input voltage V_{in} falls within a defined voltage window bounded by an upper threshold V_H and a lower threshold V_L . When V_{in} is above V_H , LED1 activates; when V_{in} is below V_L , LED2 activates; when V_{in} lies within the window ($V_L \leq V_{in} \leq V_H$), both LEDs remain OFF, indicating the in-range condition. The reference voltages are established using a precision resistor voltage divider from a regulated $\pm 12V$ dual supply. This project reinforces op-amp comparator theory and explores its practical application in voltage monitoring, industrial protection circuits, battery management systems, and analog signal conditioning.

Related Work / Literature Review

Voltage comparator circuits based on op-amps have been studied extensively in analog electronics literature. The single-threshold comparator compares an input against one reference, producing a binary output. The Window Comparator extends this by using two comparator stages simultaneously, one for the upper bound and one for the lower bound, to implement a two-sided voltage test. The IC741 is a classic general-purpose op-amp with an open-loop voltage gain of approximately 200,000 V/V (about 106 dB), making it highly effective as a comparator: even a microvolt differential input drives the output to near the supply rail. When operated open-loop without negative feedback, the IC741 exhibits output saturation at approximately $\pm(V_{supply} - 2V)$, giving approximately $\pm 10V$ for a $\pm 12V$ supply. The resistive voltage divider is a well-established method for setting precise reference voltages. LED indicators provide a simple, low-cost, visually intuitive output state display. The present design integrates these principles to build a reliable and adjustable voltage window detector using minimal components, reinforcing core concepts from 21ECC222L.

Aims and Objectives

The aim of this project is to design and implement a Window Comparator circuit that correctly identifies whether an applied input voltage lies within, above, or below a predefined voltage window using two IC741 op-amps and two red LEDs.

Objectives:

- Configure two IC741 op-amps as open-loop voltage comparators for upper (V_H) and lower (V_L) threshold detection.
- Establish V_H and V_L using a resistor voltage divider network connected to the regulated dual supply.
- Use LED1 to indicate $V_{in} > V_H$ (overvoltage) and LED2 to indicate $V_{in} < V_L$ (undervoltage); both LEDs OFF when V_{in} is within the window.
- Power the circuit using a $\pm 12V$ dual supply for proper IC741 operation and output saturation.
- Verify circuit performance by varying V_{in} across the full range and observing LED state transitions at each threshold.

Engineering Approach

The Window Comparator operates on the principle of simultaneous two-sided voltage comparison using two IC741 op-amp comparator stages.

Upper Comparator (U1 – IC741): V_{in} is applied to the non-inverting input (pin 3). The upper reference voltage V_H is applied to the inverting input (pin 2). When $V_{in} > V_H$, the differential input (pin 3 – pin 2) is positive, and the IC741 output (pin 6) saturates to $+V_{sat} \approx +10V$. This forward-biases LED1 through a 470Ω current-limiting resistor, turning it ON to indicate overvoltage.

Lower Comparator (U2 – IC741): V_{in} is applied to the inverting input (pin 2). The lower reference voltage V_L is applied to the non-inverting input (pin 3). When $V_{in} < V_L$, the differential input (pin 3 – pin 2) is positive, and U2 output saturates to $+V_{sat} \approx +10V$. This forward-biases LED2 through a 470Ω resistor, turning it ON to indicate undervoltage.

Window Condition: When $V_L \leq V_{in} \leq V_H$, U1 differential input is negative ($V_{in} < V_H$) so U1 output is $-V_{sat} \approx -10V$; U2 differential input is also negative ($V_L < V_{in}$) so U2 output is $-V_{sat} \approx -10V$. Both LEDs are reverse-biased and remain OFF, confirming the in-window condition.

Reference Voltage Calculation: Three equal resistors $R1 = R2 = R3 = 10k\Omega$ form a voltage divider from $+12V$ to ground. $V_H = (2/3) \times 12V = 8V$; $V_L = (1/3) \times 12V = 4V$. This sets a 4V-wide window centered at 6V.

LED Current: With $V_{sat} = +10V$ and $V_f(LED) \approx 2V$, the LED current $= (10 - 2) / 470 \approx 17 \text{ mA}$, which is within the safe operating range (10–20 mA) for standard red LEDs.

Course concepts applied: op-amp open-loop gain, output saturation, voltage divider analysis, LED forward voltage characteristics, and dual supply biasing. New concepts explored: window comparator topology, output logic from dual comparators, and threshold voltage engineering.

Key Components

Component	Description	Model	Qty	Function
Op-Amp	Operational Amplifier	IC741	2	Comparator stages U1 and U2
Red LED	5mm Red LED	Generic	2	Over/under-voltage indicators
Resistors	10k Ω carbon film	Generic	3	Voltage divider for V_H , V_L
Resistors	470 Ω carbon film	Generic	2	LED current limiting
Potentiometer	10k Ω	Generic	1	Adjustable V_{in} source
Breadboard	Medium size	Generic	1	Circuit assembly
$\pm 12V$ Dual Supply	Lab power supply	—	1	IC741 biasing
Jumper Wires	Connecting wires	Generic	1 set	Circuit connections

Project Deliverables

- Fully assembled and tested window comparator circuit on a breadboard demonstrating correct LED responses for all three input voltage regions.
- Complete circuit schematic showing both IC741 stages, voltage divider, LED connections with current-limiting resistors, and dual power supply connections.
- Comprehensive project report covering design theory, threshold calculations, experimental observations, and analysis.
- Demonstration showing LED state transitions at both threshold voltages V_H and V_L as V_{in} is varied.

Project Management

S.No.	Task	Duration	Start	Finish	Responsibility
1	Literature review, schematic design, threshold calculations	7 days	Week 1 Day 1	Week 1 Day 7	Suprajith
2	Component procurement and breadboard assembly	3 days	Week 2 Day 1	Week 2 Day 3	Abin Raj
3	Circuit testing and threshold calibration	5 days	Week 3 Day 1	Week 3 Day 5	Kishen Vetharth M
4	Performance verification under varying V_{in}	4 days	Week 3 Day 6	Week 4 Day 2	All Members
5	Report writing and submission	2 days	Week 4 Day 6	Week 4 Day 7	All Members

Budget

Item / Parts	Supplier	Model	Qty	Unit (Rs.)	Total (Rs.)
Operational Amplifier	Texas Instruments	IC741	2	120	240
Red LED 5mm	Generic	Red LED	2	5	10
Resistors 10k Ω	Generic	Carbon Film	3	5	15
Resistors 470 Ω	Generic	Carbon Film	2	5	10
Potentiometer 10k Ω	Generic	10K Pot	1	40	40
Breadboard (Medium)	Generic	—	1	80	80
Jumper Wires	Generic	—	1 set	100	100
Total (Rs.)					495

References

- [1] Sedra, A. S. and Smith, K. C., Microelectronic Circuits, Oxford University Press, 7th Edition, 2014.
- [2] Razavi, B., Design of Analog CMOS Integrated Circuits, McGraw-Hill, New York, 2001, pp. 220–245.
- [3] Franco, S., Design with Operational Amplifiers and Analog Integrated Circuits, McGraw-Hill, 4th Edition, 2014.
- [4] Boylestad, R. L. and Nashelsky, L., Electronic Devices and Circuit Theory, Pearson, 12th Edition, 2015.

Declaration

Project Team Member – 1	Suprajith N (RA2411053010064)
Project Team Member – 2	Abin Raj (RA2411053010069)
Project Team Member – 3	Kishen Vetharth M (RA2411053010074)
Project Supervisor	Dr E Sivakumar
Project Coordinator	Dr E Sivakumar